3.7 Other models and nonlinearities

We now compare the numerical phase diagram for our toy model to that of Anthropic's autoencoder model [EHO+22b]. The key difference between these models is that we sum the output of the nonlinear layer before computing the loss whereas in Anthropic's model the loss is computed by summing the squared differences on the output layer. This means that theirs is tasked with data compression (i.e. recovering an encoded vector) while ours is tasked with a nonlinear computation.

We compare these models across three nonlinearities: quadratic, ReLU, and GeLU. In our model when we vary the nonlinearity we do so both in the ground truth and in the activations. We further consider two cases: one with six features embedded in three dimensions ("6 in 3") and one with six features embedded in five dimensions ("6 in 5").

The top row of Figure 7 shows the results for our toy model. The same general shape is the same in all cases, as predicted from our theoretical results: a sharp 0-1 transition at mild sparsity (low kurtosis), and a smoother transition with a region of superposition for more extreme sparsity (higher kurtosis).

The superposition phase of our model also seems to be smoother, with capacity changing steadily from 0 to 1 (as predicted by our analytic results). In contrast, the Anthropic model has a large region where the capacity used by a feature is "stuck" close to 1/2. This presumably corresponds to an antipodal embedding geometry; or, in the terminology of Section 4, a separate block of the embedding matrix containing two features in a one-dimensional subspace. If the neuron basis is rotated appropriately, this becomes a single neuron that represents two features, with neither the neuron nor the features overlapping with anything else. By contrast, the superposed features in our model unavoidably project onto many neurons regardless of how the neuron basis is rotated.

Interestingly, some aspects of the phase diagram seem to depend on the nonlinearity and/or model. Our model has a narrower superposed region than Anthropic's model. For the "6 in 3" case with significantly more features than neurons, Anthropic's model shows features in superposition even when their importance is orders of magnitude larger than any other feature, which does not happen for our model (or for Anthropic's when there is only one more feature than neurons).

4 Geometry of efficient matrices

In this section we study the geometric of *efficient* embedding matrices, namely those which saturate the total capacity bound $\sum_i C_i \leq D$ with the capacity defined in equation (3). In particular, we exhaustively enumerate the forms these matrices can take and comment on the implications for the optimal capacity allocation in models.

4.1 Diagonal (monosemantic)

The simplest way to make W efficient is to put D of the vectors orthogonal to each other with arbitrary lengths and set the rest to zero. This makes W a diagonal matrix padded with zeros.

In this case, the capacities C_i are either 0 or 1. The norms-squared n_i are zero (if $C_i = 0$) or nonzero but arbitrary (if $C_i = 1$).

4.2 Semiorthogonal ("everything bagel")

At the other extreme, we can choose $W = \sqrt{\lambda}R$ where R is semiorthogonal, meaning that $RR^T = I$. (This means the rows of W are orthonormal, but the columns aren't.)

In this case, we have $WW^T = \lambda I$, so

$$(W^TW)^2 = \lambda W^TW \tag{21}$$

and the capacity simplifies to

$$C_i = \frac{1}{\lambda}[W^TW]_{ii} = \frac{n_i}{\lambda} \tag{22}$$

So the capacities and norms of vectors are directly related.

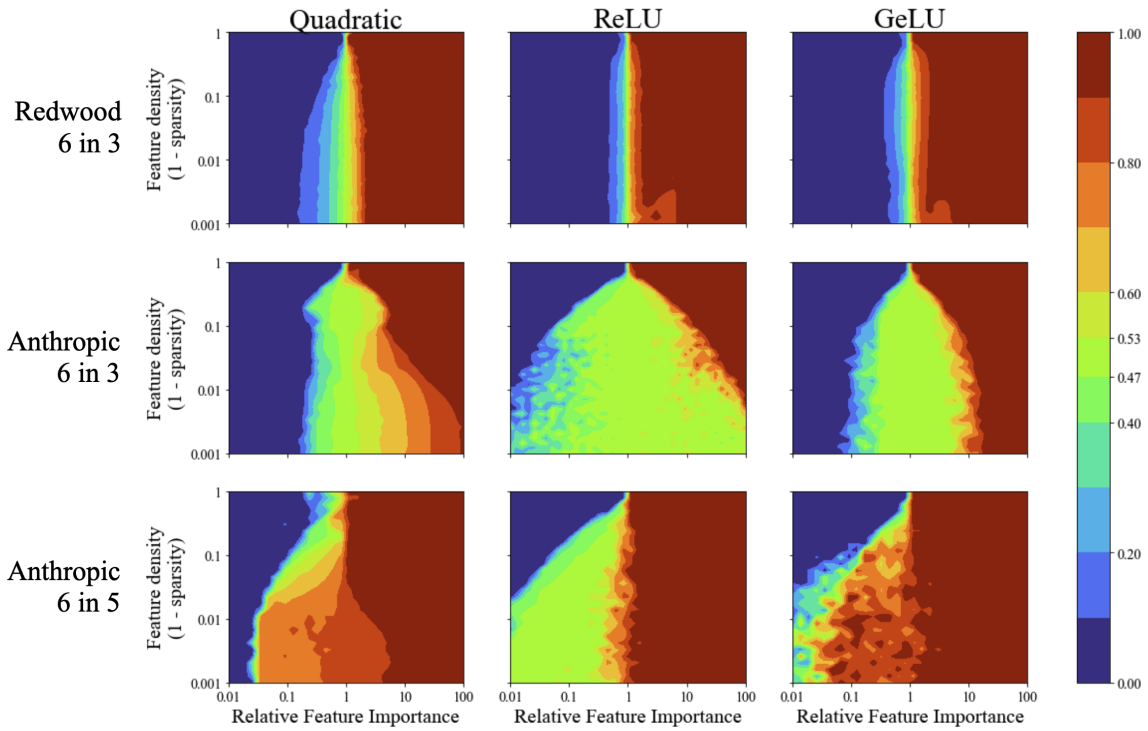

Figure 7: Empirical phase diagrams are shown for our toy model, Anthropic's toy model with 6 features in 3 dimensions, and Anthropic's toy model with 6 features in 5 dimensions. For each model we study three different nonlinearities. Our toy model looks similar for 6 features in 3 dimensions and 6 features in 5 dimensions for all nonlinearities, so we only display plots of 6 features in 3 dimensions. The contour at $c = 0.5$ has been replaced with two contours at $c = 0.47, 0.53$ to give more detail for the large $c \approx 1/2$ regions, which correspond to the antipode polytope from [EHO+22b].

We now confirm that this is efficient:

$$\sum_i C_i = \frac{1}{\lambda}\text{Tr}(W^T W) = \frac{1}{\lambda}\text{Tr}(WW^T) = \frac{1}{\lambda}\lambda D = D \tag{23}$$

Theorem 1 *Such a matrix exists for* any $\vec{C}$ *that sums to* D*, so this gives us a way of dividing up capacity efficiently however we like.*

The proof is shown in Appendix H.

The singular values of a semiorthogonal matrix are all equal (and vice versa), so the condition of semiorthogonality is equivalent to requiring embedding vectors to be distributed in all directions in a balanced, isotropic way.[5] This gives some geometric intuition for what the angles between vectors are doing: shorter vectors, for features with less capacity, will be spaced closer together in order to keep the overall arrangement balanced.

4.3 Mixed (tegum product)

We can combine these approaches by splitting our embedding space into orthogonal subspaces of dimension D_k and using a semiorthogonal matrix to embed N_k of our vectors into the kth subspace.

To do this, make W block-diagonal with blocks $\lambda_k R_k$, where R_k is a $D_k \times N_k$ semiorthogonal matrix. Here $\sum_k D_k = D$ and $\sum_k N_k = N$. For an example with two blocks, see Figure 5.

[5] The exact sense of "balanced" is a little tricky: the vectors don't have to be centered around the origin! Instead, think about a dataset whose principal components all have the same variance.

Within each block, the dimensions of a subspace are divided up as before:

$$n_i = \lambda_k C_i \quad \text{if } i \in \text{block } k \tag{24}$$

$$\sum_{i \in \text{ block } k} C_i = D_k \tag{25}$$

Note that the independent scalars λ_k let us scale different subspaces arbitrarily. (This is the only thing that makes this more general than the previous case.)

4.4 General

So far we have chosen subspaces aligned with the standard basis on $\mathbb{R}^D$ (i.e. the neurons), but we can rotate our set of embedding vectors however we like without changing $\vec{C}$ or $\vec{n}$. Similarly, we have assigned dimensions of $\mathbb{R}^N$ (i.e. features) to subspaces sequentially, but we can do it in any order.

In other words: a sufficient condition for efficiency is that W is a $D \times N$ matrix of the form QBP, where

- Q is a $D \times D$ orthogonal matrix,
- P is an $N \times N$ permutation matrix, and
- B is a $D \times N$ block-diagonal matrix whose (rectangular) blocks are of the form $R_k\sqrt{\gamma_k}$, where γ_k is a positive scalar and R_k is a semiorthogonal matrix.

In terms of the SVD decomposition $W = QSR$, an equivalent condition is that rows of R corresponding to unequal singular values S_{ii} are never nonzero in the same column.

Theorem 2 *This is also a necessary condition; all efficient matrices are of this form.*

The proof is shown in Appendix H.

4.5 Interpolating between regimes

Consider some block of an efficient matrix (factor of a tegum product), and some vector $\vec{w}_i$ in that block. If the norm-squared n_i of that vector is increased, its capacity C_i increases in order to keep $c \propto n$ within the block. At the same time, the vector becomes more orthogonal to the rest of the block.

If the capacity reaches $C_i = 1$, the vector becomes fully orthogonal and separates into a new block. Once that happens, n_i can be varied arbitrarily without affecting C_i.

On the other hand, if the capacity decreases to $C_i = 0$, then $n_i = 0$ and the vector becomes zero; at that point the feature is ignored.

4.6 Application to toy models

Empirically, Anthropic found that (when importances are equal) their embedding matrices often factor into orthogonal subspaces containing simple geometric structures [EHO+22b]. This is equivalent to the statement that their embedding matrices have a large number of small blocks. (This also explains some features of the phase diagram for their model, such as a large region near $c = 1/2$; see above.) In contrast, the embedding matrices in our model have one large superposed block (except for ignored or fully-represented features).

We can understand this phenomenon by thinking about the constraints on $\vec{C}$ and $\vec{n}$ in the large-block and small-block regime. When there is one large block, capacity can be divided up arbitrarily; however, the lengths of embedding vectors are determined entirely by the capacities (up to an overall scalar). For our model, this is compatible with good loss, because minimizing loss gives the relationship $C_i \propto n_i$ anyway (see above). However, for Anthropic's model, this may be more constraining of embedding vector norms than the loss function "wants".

In contrast, a collection of small blocks imposes constraints on capacity allocation, because each block must separately add up to the correct capacity. Once capacity is allocated, however, the lengths of vectors can be chosen much more freely by scaling blocks independently of each other. This doesn't provide any benefit for our toy model, but seems to be useful to Anthropic's.